

SEEING TROUBLE BEFORE IT FAILS

AI-BASED INSULATOR HEALTH DETECTION **FOR SPOT** **MAINTENANCE**

Custom YOLO object detection on a Roboflow dataset —
moving **EHT & HT networks** from reactive inspection to
condition-based maintenance.

DRONE IMAGERY · ROBOFLOW ANNOTATION · ULTRALYTICS YOLO · SPOT DISPATCH

Illustrative inspection imagery

THE SILENT BACKBONE FAILS QUIETLY

Insulators are the silent backbone of **EHT & HT transmission and distribution** — they provide electrical insulation and mechanically support live conductors.

- Mounted **high on towers**, in hard-to-reach locations and difficult terrain.
- Manual inspection is **time-consuming, resource-intensive and hazardous** for field crews.
- Visible deterioration goes **unnoticed until failure** → flashovers, outages, reactive emergency maintenance.



**DAMAGE IS VISIBLE —
NOBODY IS LOOKING**

Illustrative inspection imagery

EYES ON EVERY TOWER — FEET ON THE GROUND

Drones / UAVs with high-resolution cameras capture **images and video of installed insulators safely from close range** — covering heights and difficult terrain in a fraction of the time, and producing the visual datasets that make AI inspection possible.

Illustrative inspection imagery

01

PERSONNEL SAFETY

No climbing live towers, no risky ground patrols in rough terrain.

02

INSPECTION SPEED

Cover spans and heights in a fraction of the time a crew needs.

03

REPEATABLE FLIGHTS

Same route, same angles — condition compared cycle over cycle.

04

DIGITAL RECORDS

Every asset photographed, dated and archived — a real audit trail.

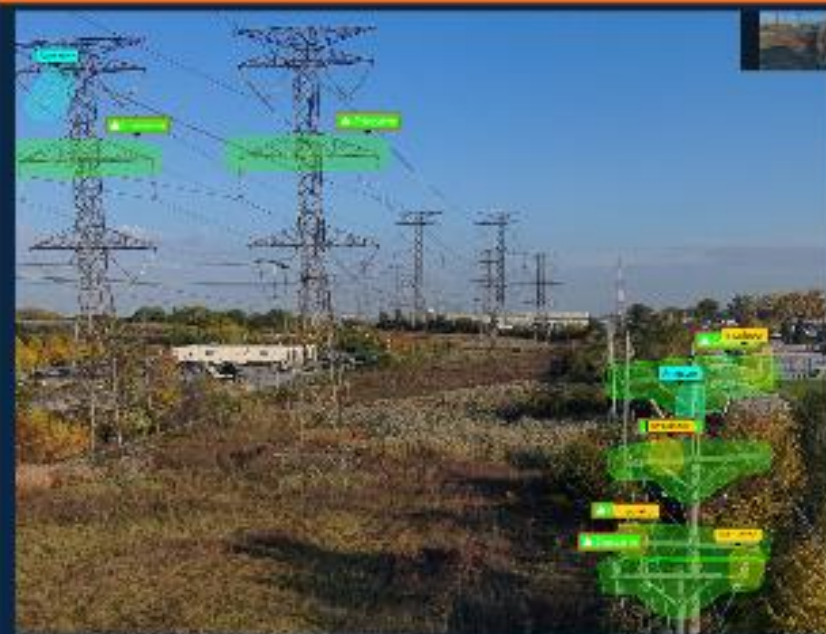
A MODEL THAT READS EVERY FRAME

A **custom YOLO deep-learning model**, trained on annotated Roboflow data, automatically detects and classifies visible insulator conditions in images and video.

OUTPUT

Per-image detections — **healthy / broken / flashover ...** — each with a bounding box and a confidence score.

These detections power **spot-maintenance planning**: crews prioritise the assets needing inspection, repair or replacement **before they become major failures**.



DETECTION, NOT GUESSWORK

Boxes and labels on every asset in frame, ranked by confidence.

Illustrative inspection imagery

ANNOTATED IMAGERY IS THE REAL ASSET

Fully annotated insulator imagery. Typical published classes are **"insulator"**, **"broken insulator"** and **"pollution-flashover"** — Kaggle's Insulator Defect Detection defines 3 sub-classes: flashover damage and broken insulator shell.

INSULATOR

Healthy · baseline class

BROKEN

Shell damage · fracture

FLASHOVER

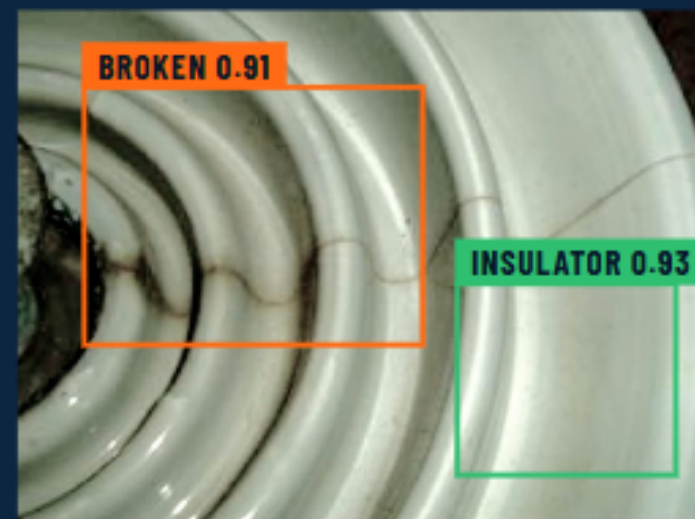
Pollution · surface arcing

Roboflow provides **ready-to-train exports** with train / validation / test split and augmentation built in.

OUR PROJECT DATASET

DATASET SIZE & CLASS COUNTS: TO BE MEASURED

Sources: kaggle.com/competitions/insulator-defect-detection · universe.roboflow.com/use-cases/insulator-defect-detection



ANNOTATION STYLE

Every defect boxed and class-labelled — the ground truth YOLO learns from.

Illustrative inspection imagery; boxes shown for annotation style only

FROM FLIGHT TO WORK ORDER

Seven stages, one continuous loop — imagery in, prioritised dispatch out.

01

CAPTURING

Drone imagery of
EHT / HT assets

02

ANNOTATING

Roboflow class
labelling

03

TRAINING

Ultralytics YOLO on
GPU

04

VALIDATING

Held-out test
images

05

INFERENCE

Boxes + confidence
scores

06

FLAGGING

Defects raised for
action

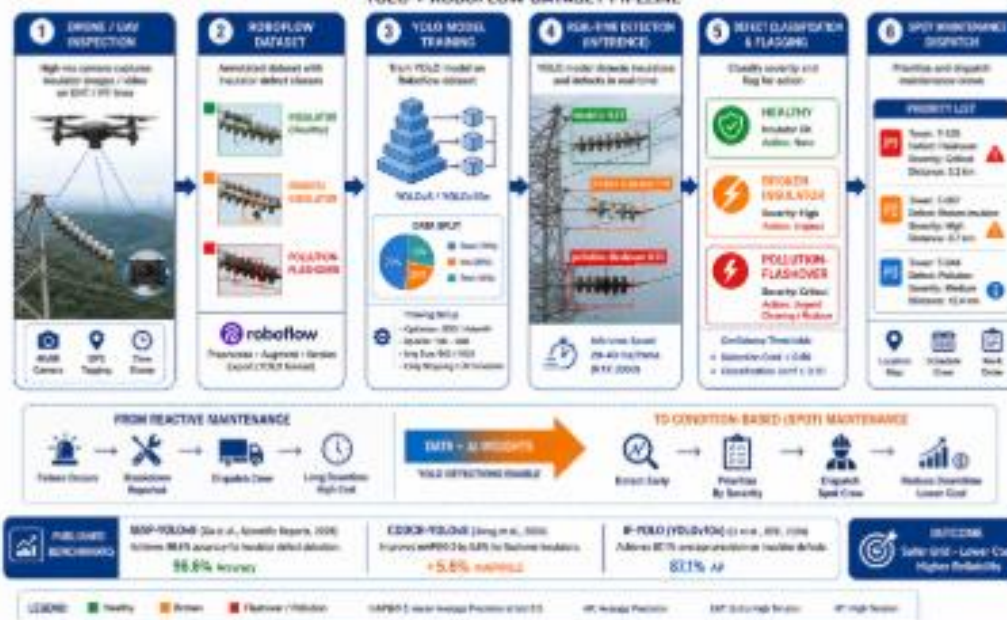
07

DISPATCH

Spot maintenance
with priorities

AI-BASED INSULATOR HEALTH DETECTION FOR SPOT MAINTENANCE

YOLO + ROBOFLOW DATASET PIPELINE



PROVEN ON THIS EXACT PROBLEM

Single-stage, real-time detection — fast enough to analyse video frames streaming from drones.

Mature **Ultralytics ecosystem** — train, tune, export and deploy with a supported toolchain.

PUBLISHED EVIDENCE THIS WORKS

MAP-YOLOV8

96.6%

average accuracy for insulator defect detection — **14% higher** than baseline YOLOv8

Xu et al., Scientific Reports 2025
nature.com/articles/s41598-025-92445-3

CDDCR-YOLOV8

+5.6%

improvement in mAP@0.5 for flashover insulator prediction

Jiang et al. 2024
link.springer.com/article/10.1007/s44196-024-00654-x

IF-YOLO

(YOLOV10N-BASED)

87.1%

average precision on insulator defects

Li et al., IEEE Access 2024
ieeexplore.ieee.org/iel8/6287639/10380310/10750545.pdf

YOLOV7 VS YOLOV8

Benchmarked

direct comparison for broken insulator detection

Fahim et al. 2024
sciencedirect.com/science/article/pii/S2772671124002432

These are **published results from other studies** — they establish feasibility, not our outcome.

MEASURED, NOT CLAIMED

- **Ultralytics YOLO training loop with GPU** acceleration.
- Metrics tracked: **precision, recall, mAP@0.5, mAP@0.5:0.95, F1-score.**
- **Augmentation from Roboflow** improves robustness to lighting, weather and viewing angle.
- **Small-object handling tuned** for distant insulators high on towers.

PROJECT RESULTS

Our own model has not been evaluated yet — no figures are stated.

METRIC	VALUE
Precision	TO BE MEASURED
Recall	TO BE MEASURED
mAP@0.5	TO BE MEASURED
mAP@0.5:0.95	TO BE MEASURED

Figures will be populated from the held-out validation run in Phase 2.

FIX EXACTLY WHAT NEEDS FIXING

A detection is only useful once it becomes a work order.

STEP 01

INFER

Run YOLO over new drone imagery

STEP 02

CLASSIFY

Assign condition per insulator

STEP 03

RANK

Severity / priority P1 → P3

STEP 04

DISPATCH

Crew sent with coordinates + photos

STEP 05

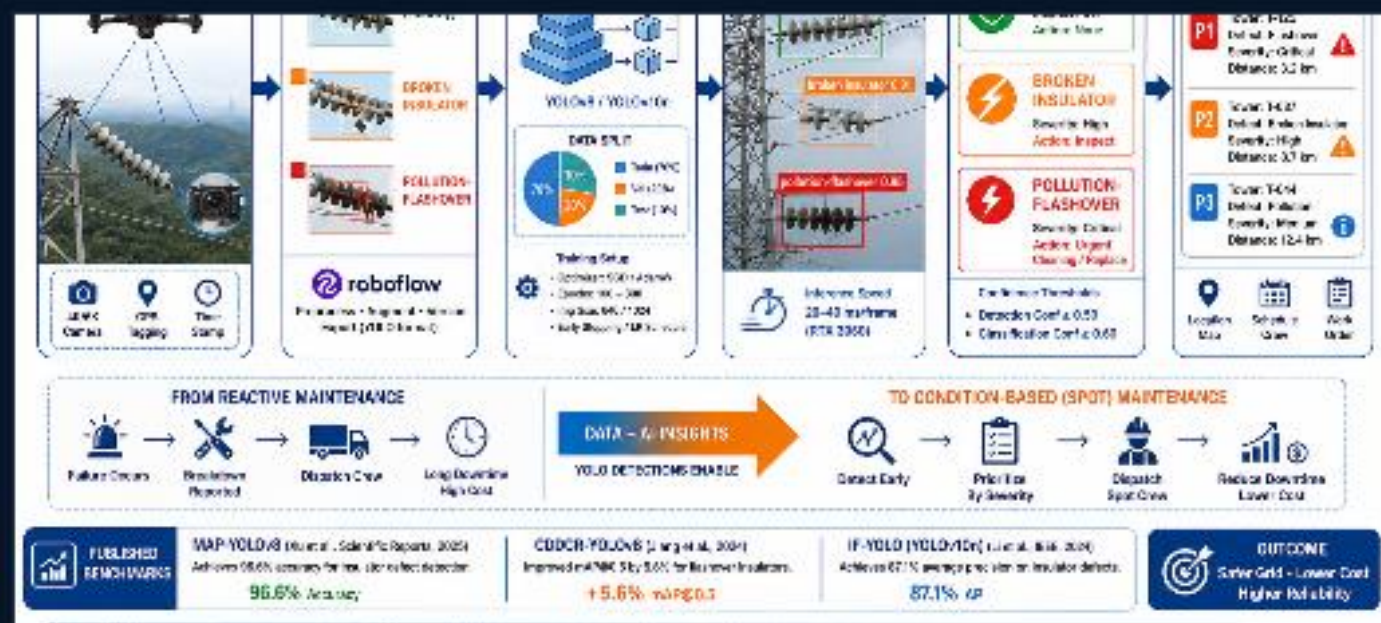
REPAIR

Fix or replace before failure

STEP 06

UPDATE

Asset records refreshed



Workflow reference diagram - illustrative inspection imagery

THIS IS SPOT MAINTENANCE

FIX EXACTLY WHAT NEEDS FIXING, WHEN IT NEEDS FIXING.

No blanket patrols. No emergency call-outs at 2 a.m. Maintenance effort follows real, evidenced asset condition — **P1 critical first, P3 minor when convenient.**

WHAT THE GRID GETS

Safety, speed and spend — all improved by knowing asset condition earlier.

01

IMPROVED CREW SAFETY

Less climbing on live structures, fewer ground patrols across difficult terrain.

02

FASTER INSPECTION CYCLES

More spans covered per shift; automated review instead of manual photo triage.

03

EARLIER DEFECT DETECTION

Visible deterioration caught while it is still a repair, not an outage.

04

SPEND FOLLOWS CONDITION

Maintenance budget prioritised by real asset condition, not by calendar rota.

05

FEWER UNPLANNED OUTAGES

Fewer flashover events and emergency interruptions on EHT & HT feeders.

06

DIGITAL AUDIT TRAIL

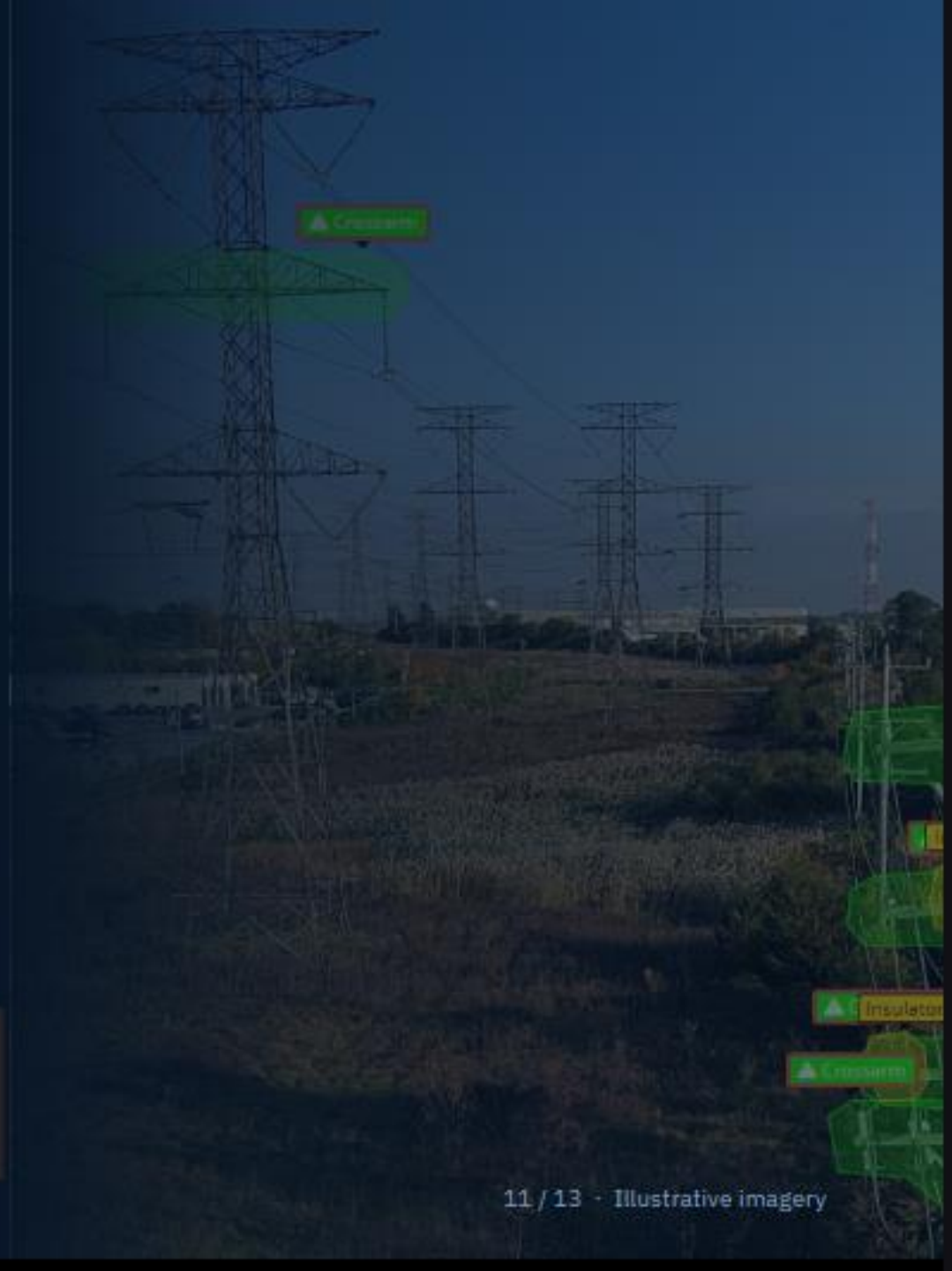
Dated, geotagged evidence per asset — defensible inspection history.

FROM DETECTION TO PREDICTION

- 01 **Severity classification** of each detected defect, not just its class.
- 02 **Asset-level risk scoring** that combines condition, criticality and history.
- 03 **GIS-based visualisation** of network health across the whole corridor.
- 04 **Automated inspection reports** generated straight from detections.
- 05 **Integration** with utility maintenance and asset-management systems.

THE ENDGAME

**AN AI-POWERED PREDICTIVE ASSET-HEALTH MONITORING
PLATFORM FOR THE WHOLE T&D NETWORK**



FIVE PHASES TO SCALE

Each phase produces a working artefact — dataset, model, pilot, workflow, platform.



THROUGH-LINE

REACTIVE INSPECTION → CONDITION-BASED MAINTENANCE

SEEING TROUBLE BEFORE IT FAILS

SEE IT EARLY. FIX IT SPOT-ON. KEEP THE GRID LIVE.

THANK YOU • QUESTIONS

Drone imagery • Roboflow annotation • Ultralytics YOLO • prioritised spot maintenance for EHT & HT networks.

Illustrative inspection imagery

REFERENCES

Xu et al., *Scientific Reports* 2025 — MAP-YOLOv8, 96.6% average accuracy, 14% higher than baseline YOLOv8 · nature.com/articles/s41598-025-92445-3 | Jiang et al. 2024 — CDDCR-YOLOv8, mAP@0.5 improved by 5.6% for flashover insulators · link.springer.com/article/10.1007/s44196-024-00654-x
Li et al., *IEEE Access* 2024 — IF-YOLO (YOLOv10n-based), 87.1% average precision · ieeexplore.ieee.org/iel8/6287639/10380310/10750545.pdf | Fahim et al. 2024 — YOLOv7 vs YOLOv8 for broken insulator detection · sciencedirect.com/science/article/pii/S2772671124002432